

Data Science & AI Bootcamp – Interview Question Bank

MATH AND STATISTICS

1. Explain probability distributions.
2. What is hypothesis testing?
3. Explain confidence intervals.
4. What is correlation and covariance?
5. How do you calculate p-values?
6. What is standard deviation?
7. Discuss a few techniques for handling outliers.
8. Explain linear algebra basics.
9. What is matrix multiplication?
10. How do you calculate derivatives?
11. Explain basic probability concepts.
12. What is Bayes' theorem?
13. How do you calculate variance?
14. Explain sampling methods.
15. What is the central limit theorem?
16. How do you handle missing data?
17. Explain regression analysis.
18. What is statistical significance?
19. How do you calculate sample size?
20. Explain different statistical tests.
21. Explain different distance metrics.
22. What is dimensionality reduction?
23. How do you handle multivariate statistics?
24. What are different probability distributions?
25. How do you implement hypothesis testing?
26. Explain Bayesian statistics.
27. What are different sampling techniques?
28. Explain different regression techniques.
29. What are different optimization methods?
30. Explain different clustering metrics.
31. What are different estimation techniques?
32. How do you handle statistical inference?
33. Explain different statistical models.
34. What are different statistical learning methods?
35. What Are the Main Measures Used to Describe the Central Tendency of Data?
36. What Are the Main Measures of Variability?

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- 37. What Are Skewness and Kurtosis?
- 38. What is Bayes' Theorem and when is it used in data science?
- 39. Explain the difference between a null and alternative hypothesis.
- 40. How do you handle situations where the data does not meet the assumptions of a hypothesis test?